

IB MATH AA SL

Lesson 21

Sine & Cosine Rules

Trigonometry · 60-minute lesson

Syllabus SL 3.2–3.3 · Property of Lynda Badmus Education



Do Now — recall

ON YOUR WHITEBOARD

1. In a right triangle, what is $\sin = ?$
2. Sum of angles in any triangle?
3. Which rule needs a right angle: Pythagoras or sine rule?

Answers (click to reveal)

1. opposite/hypotenuse
2. 180°
3. Pythagoras

What we're learning today

LEARNING OBJECTIVES

- 1** **Use**
the sine rule to find sides and angles.
- 2** **Use**
the cosine rule to find sides and angles.
- 3** **Choose**
the correct rule for the information given.
- 4** **Find**
the area with $\frac{1}{2}ab \sin C$.

SUCCESS CRITERIA — I CAN...

- ✓
apply the sine rule when an angle–opposite-side pair is known
- ✓
apply the cosine rule for SAS or three sides
- ✓
decide which rule a problem needs
- ✓
find a triangle's area with $\frac{1}{2}ab \sin C$

The sine rule

PAIR UP SIDE & ANGLE

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Use when

you know an angle and its opposite side

Sides

lower-case; opposite angles upper-case

WHICH RULE?

Sine rule

a side & its opposite angle known

Cosine rule

two sides + included angle, or all three sides

Sine rule

QUESTION

In triangle ABC, $A = 40^\circ$, $B = 60^\circ$, $a = 8$ cm.

Find side b .

SOLUTION

Sine rule, pairing $a/\sin A$ with $b/\sin B$:

$$\frac{b}{\sin 60^\circ} = \frac{8}{\sin 40^\circ}$$

Make b the subject:

$$b = \frac{8 \sin 60^\circ}{\sin 40^\circ} \approx 10.8 \text{ cm}$$

Your turn — sine rule

ANSWER IN YOUR BOOK

1. $A=50^\circ$, $a=10$, $B=30^\circ$. Find b .

2. $A=70^\circ$, $a=12$, $C=40^\circ$. Find c .

(Give answers to 3 s.f.)

Answers (click to reveal)

1. $b=10 \sin 30 / \sin 50 \approx 6.53$

2. $c=12 \sin 40 / \sin 70 \approx 8.21$

The cosine rule

FOR SAS OR SSS

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Use when

you know two sides and the included angle

Or

all three sides (rearrange for the angle)

AREA

$$\text{Area} = \frac{1}{2} ab \sin C$$

uses two sides and the angle between them.

Cosine rule

QUESTION

In triangle ABC, $b = 7$, $c = 9$, $A = 50^\circ$.

Find side a .

SOLUTION

Cosine rule with the angle A between b and c :

$$a^2 = 7^2 + 9^2 - 2(7)(9)\cos 50^\circ$$

Evaluate: $49 + 81 - 126\cos 50^\circ \approx 49.0$:

$$a \approx 7.00 \text{ cm}$$

Exam-style question

In triangle ABC, $a = 11$ cm, $b = 8$ cm and $C = 35^\circ$.

- (a) Find side c . [3]
- (b) Find the area of the triangle. [2]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $c^2 = 11^2 + 8^2 - 2(11)(8)\cos 35^\circ$ (M1)(A1) $\rightarrow c \approx 6.40$ cm (A1)

(b) Area = $\frac{1}{2}(11)(8)\sin 35^\circ$ (M1) ≈ 25.2 cm² (A1)

Common mistakes



Wrong rule

Sine rule needs a side & its opposite angle; otherwise use cosine.



Label matching

Side a is opposite angle A — keep them paired.



Cosine rule sign

It is $- 2bc \cos A$ (a minus).



Area formula

$\frac{1}{2}ab \sin C$ uses the angle BETWEEN the two sides.

Mixed practice

QUESTIONS

1. $A=45^\circ$, $a=14$, $B=60^\circ$. Find b .
2. $b=5$, $c=6$, $A=80^\circ$. Find a .
3. Area: $a=9$, $b=10$, $C=30^\circ$.
4. Which rule for SSS (3 sides)?

Answers (click to reveal)

1. $14 \sin 60 / \sin 45 \approx 17.1$
2. $a^2 = 25 + 36 - 60 \cos 80 \rightarrow a \approx 7.10$
3. $\frac{1}{2} \cdot 9 \cdot 10 \cdot \sin 30 = 22.5$
4. cosine rule

Plenary & exit ticket

KEY FORMULAS

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

EXIT TICKET — before you leave

1. When do you use the sine rule?
2. Write the area formula.
3. Cosine rule for a^2 ?

Answers (click to reveal)

1. side & opposite angle known
2. $\frac{1}{2}ab \sin C$
3. $b^2 + c^2 - 2bc \cos A$

Self-assess:
 • need more help
 • nearly there
 • confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Sine & cosine rules.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 22

Modelling with sinusoidal functions — waves in the real world.

$$y = a \sin (b(x - c)) + d$$